

Renal regulation of sodium and water balance

Both water and sodium are freely filterable at the glomerulus and approximately 99% are reabsorbed in the renal tubules.

The reabsorption of **water and chloride** occur by **passive diffusion** and depends upon the **active re-absorption of sodium ions**

Renal regulation of sodium and water balance

The sodium ions are positively charged ions and when **actively re-absorbed** from the renal tubules to the blood they take the **negatively charged Cl ions passively**.

When Na and Cl ions are reabsorbed they form **NaCl** and the **osmotic pressure** becomes **higher** and subsequently a **water concentration difference** is created between the tubular lumen and the peritubular capillaries, which produce a **driving force for water re-absorption by osmosis**.

Renal tubular lumen

Peritubular capillaries

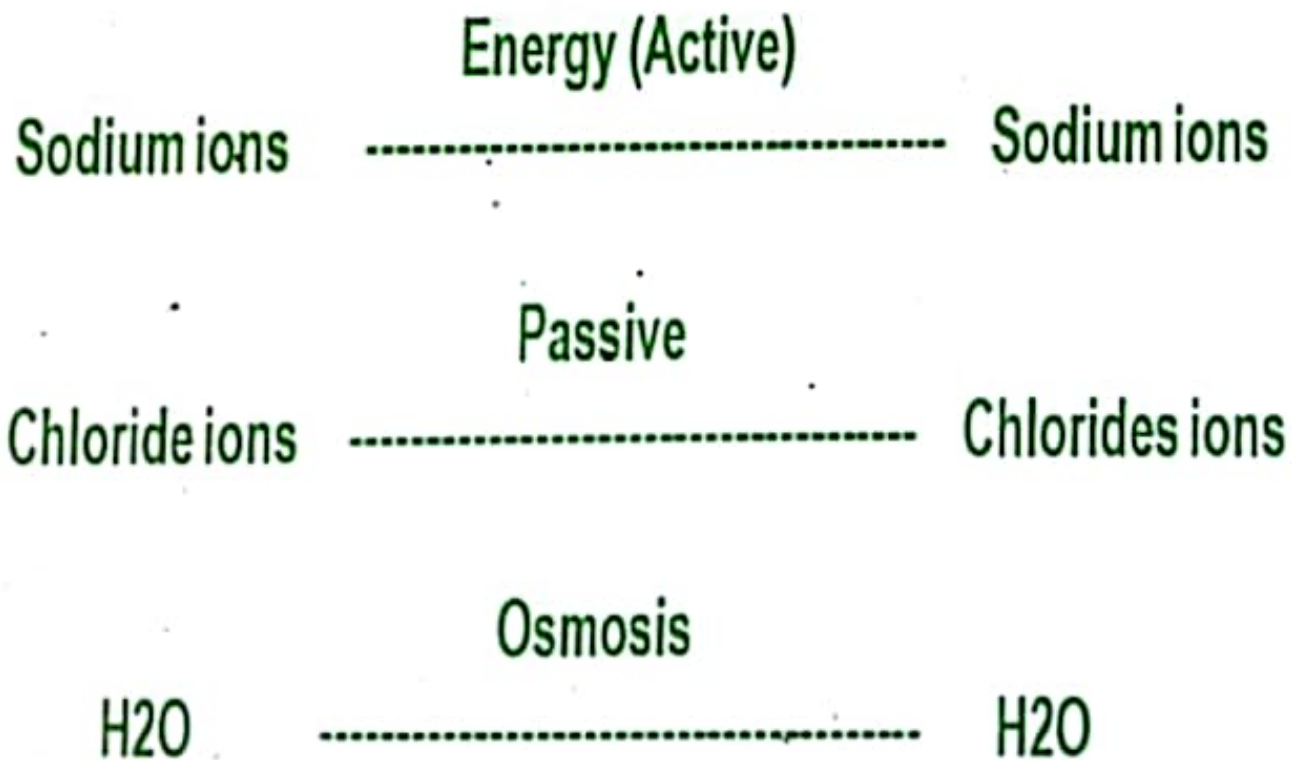


Table 2. Normal routes of water intake and loss.

Water intake		Water output	
Drinking	1200 ml/day	Insensible loss	900 ml
In food	1000 ml	Sweat	50 ml
Metabolic water	350 ml	In feces	100 m
		In urine	1500 ml
Total 2550 ml		Total 2550 ml/day	

Note

The kidney can regulate the amount of water excreted in urine from 400 ml water consumption if a person lost in a desert

or 25 liters /day if a person participating in a beer-drinking contest or a person consuming a lot of water in a war in summer

Note

The permeability of the **distal convoluted tubules and collecting tubules** to the **water** is completely under the control of **anti-diuretic hormone (ADH)** from the posterior pituitary gland.

The mechanism by which **the ADH** increase water re-absorption through increasing **the activity of cyclic AMP** in the tubular cells and subsequently, increase the number of **aquaporins channels** (water channels) and hence increasing water re-absorption.

Control of sodium excretion

The kidney can regulate the increase in body sodium by increasing its excretion in urine.

Sodium is **freely filterable** at the glomerulus and then **actively reabsorbed** but **not secreted** by the tubules.

The amount of excreted sodium could be represented by the following formula

$\text{Na excretion} = \text{Na filtered} - \text{Na reabsorbed}$

Control of sodium excretion

All mechanisms which **maintain total body sodium constant** simultaneously keep **extracellular volume constant**.

If a large quantity of sodium and water lost from the body due to diarrhea

- 1- the plasma volume will decrease
- 2- the blood pressure will also decrease
- 3- The glomerular capillary pressure will decrease
- 4- The glomerular filtration will decrease
- 5- The sodium excretion will decrease.

The reverse mechanism will occur if the sodium intake and water intake are increased.

Control of Tubular sodium reabsorption

Sodium reabsorption is completely under the control of **aldosterone hormone** secreted from the adrenal gland.

This mechanism is called **Renin-angiotensin-aldosterone mechanism**.

This mechanism started with **the decrease in extracellular volume** which will **increase rennin** secretion from the **juxtaglomerular apparatus** and

Control of Tubular sodium reabsorption

the angiotensin from the liver and finally **aldosterone** from the adrenal gland which **increase the tubular sodium reabsorption** and decrease sodium excretion.

- 1 - Decrease extracellular volume
- 2 - Increase rennin secretion from the kidney
- 3 - Increase angiotensin from the liver
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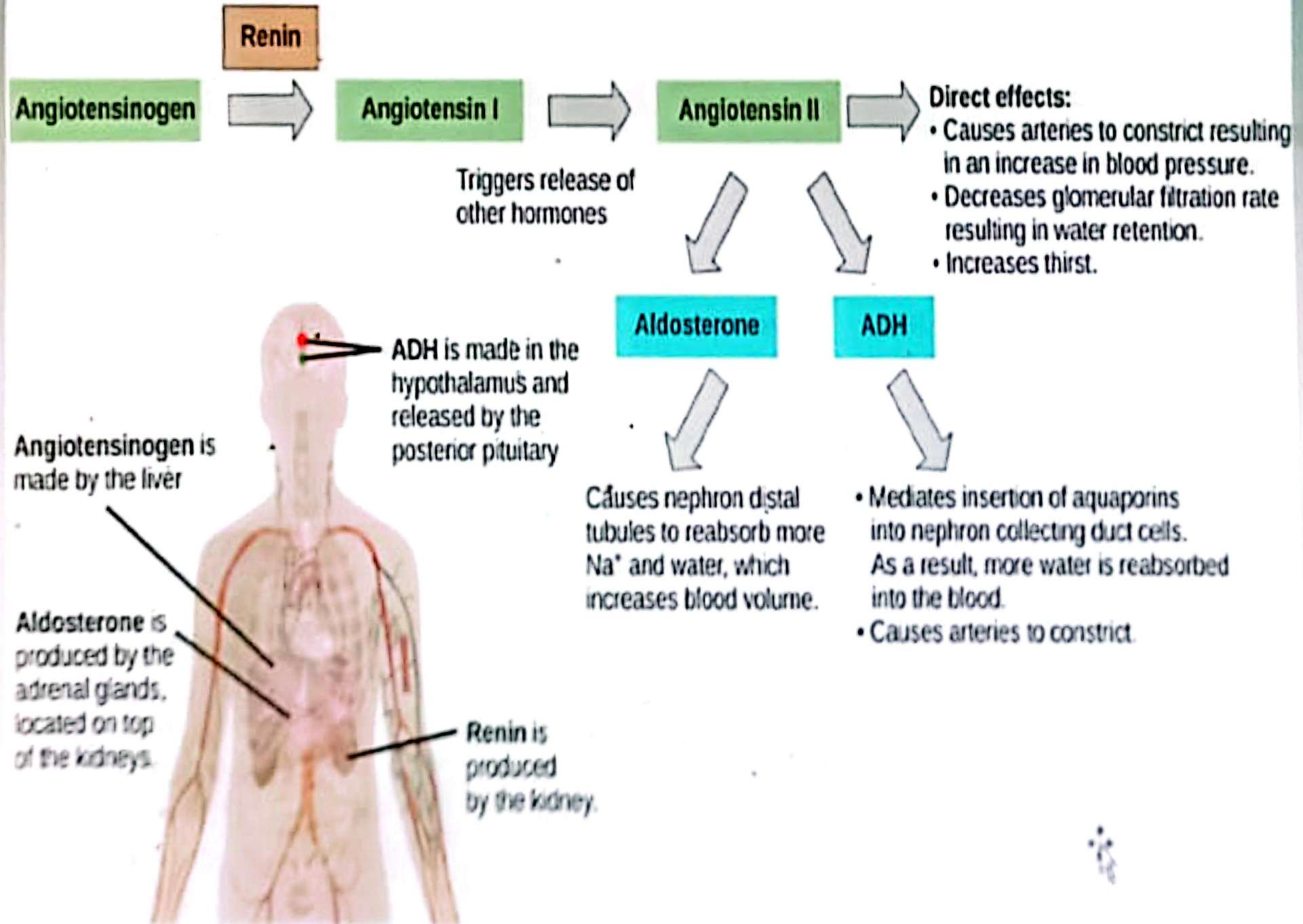
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The renin-angiotensin-aldosterone system increases blood volume and pressure



Renal regulation of extracellular volume

When the extracellular volume decreased, there are two important mechanisms occur to retain the volume to its original normal physiological level.

1- The first mechanism is the stimulation of hypothalamic neurons to produce ADH, which stimulate the renal tubules to increase water reabsorption.

2- The second mechanism by which water is regulated is through the renin-angiotensin-aldosterone mechanism

Renal regulation of extracellular volume

When the water level decrease, the following sequences will occur:

- 1- Decrease plasma volume
- 2- Decrease venous return
- 3- Decrease arterial blood pressure
- 4- Increase ADH secretion
- 5- Increase water reabsorption from renal tubules
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CONTROL OF ADH secretion

There are many **osmoreceptors** located in the hypothalamus. These receptors are **sensitive to the changes of the osmolarity** of the extracellular fluid by unknown mechanism.

An **increase in osmolarity (hypertonic)** stimulate the hypothalamic cells that **secrete the ADH** and conversely, a decrease in osmolarity inhibits ADH secretion.

So, the **ADH secretion rate** could be affected by **baroreceptors** and **osmoreceptors**.

CONTROL OF ADH secretion

Thus, an **increase in extracellular volume** and a **decrease in the osmolarity** will cause a **maximal inhibition of ADH** secretion and the opposite changes. i.e. **decrease extracellular volume** and **increase in osmolarity**, will produce **maximal stimulation of ADH**.

NB: Syndrome of inappropriate ADH (SIADH), a condition in which body makes too much ADH, and so, retains too much water.

NB. Baroreceptors: type of mechanoreceptors relaying information derived from blood pressure within autonomic N.S. that subsequently alter total peripheral resistance and cardiac output to keep blood pressure in normal range.

Notes

Alcohol is a powerful inhibitor for ADH release and account for the large volume of urine flow accompanying alcoholic drinks.

Diarrhea may induce loss of 3 liters of water and salts and decrease both the extracellular volume and the osmolarity and subsequently increase the ADH secretion.

In case of a disease called diabetes insipidus (complete absence or decrease of ADH), a large volume of highly diluted urine (25 liters/day) is excreted.

NOTES

In this case the kidney lost its ability to reabsorb water regardless of the change in extracellular volume and osmolarity.

There are two types of DI

1. **Central diabetes insipidus** (Hypothalamic – neurohypophyseal lesions), most cases due to damage to hypothalamus or pituitary, or abnormality in gene responsible for vasopressin secretion

2. **Nephrogenic diabetes insipidus** (Insensitivity of the kidney to ADH)

patients of these disorders if untreated, have the predictable result of dehydration, hyperosmolarity,

hypovolemia and eventual death in severe

Thirst

NOTES

The phenomenon of thirst is important for regulation of body water, sodium and osmolarity. Thirst, the primary regulator of water intake is defined as the conscious desire for water (drinking).

There are 2 kinds of thirst

1. Osmotic Thirst: from eating salty food & increased blood osmolarity $\longrightarrow$ animals have a preference for drink pure water.

2. Hypovolemic T.: from loss of fluid due to bleeding or sweating $\longrightarrow$ animals have a preference for slightly salty water

The centers which mediate thirst are located in the **hypothalamus** very close to the area which produce the **ADH (Supraoptic nucleus)**.

Stimulation of thirst center induces profound and prolonged drinking. There are several stimuli that lead to thirst.

1 - Angiotensin. Excess angiotensin II increase rate of drinking twice or three fold

2 - Hemorrhage and low cardiac output. Loss of 10% of blood induce thirst.

3 - Dryness of the mouth. Might be a factor for sensation of thirst.

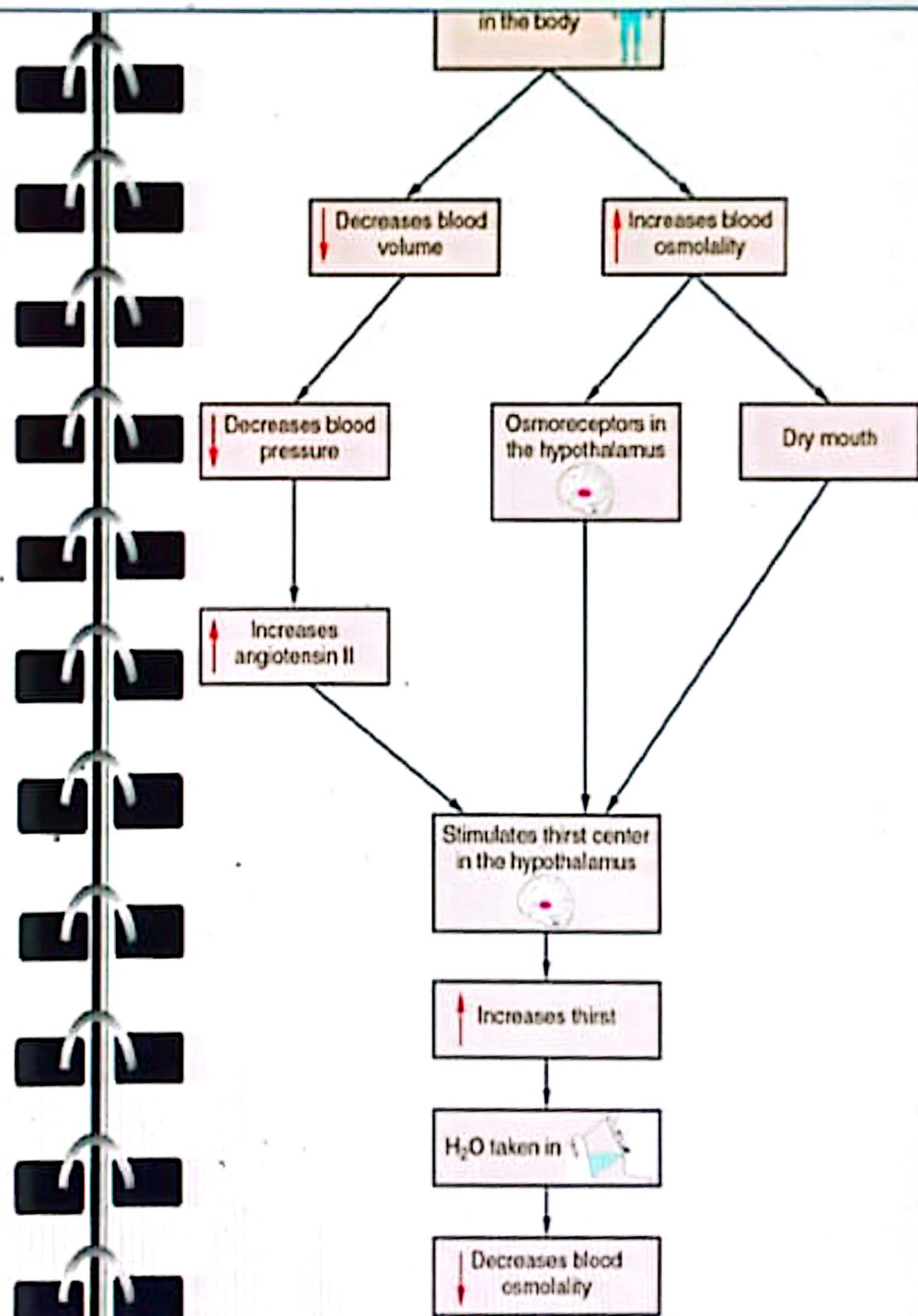
Thirst

Animals, such as camel which have been markedly dehydrated will rapidly drink enough water to replace their previous loss and then stop drinking.

It is amazing that when they stop drinking, the water has not yet had time to be absorbed from the gastrointestinal tract into the blood.

Some kind of "metering" of water intake by the gastrointestinal tract has occurred.

Note: The sensation of thirst usually decreased with advancing age. So in aged persons we should allow them to drink frequently to avoid dehydration.



Renal regulation of potassium ions

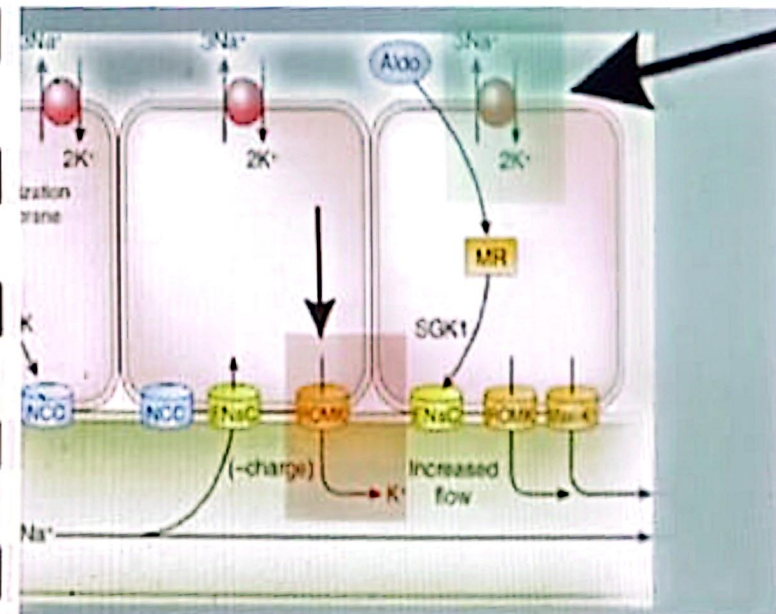
Potassium ions are important in the excitability of nerves and muscles. Most of the body K is found within the cells as a result of active Na and K pump.

All filtered K in the glomerulus is reabsorbed and the K excreted in the urine comes from tubular secretion (late DCT).

The pathway by which an increase in K intake induces greater K excretion in the urine is as follows.

Renal regulation of potassium ions

- 1- Increased K intake resulted in
- 2- Increased plasma K concentration
- 3- Direct effect on the adrenal cortex to increase aldosterone hormone secretion from zona glomerulosa
- 4- Increased K secretion from the peritubular capillaries to tubular lumen
- 5- Increased K excretion in the urine



Renal regulation of pH

Most metabolic reactions in the body are sensitive to the hydrogen-ions concentration in the media in which they occur.

The H^+ influence the enzyme function and activity.

Any compound liberating H^+ is called an ACID and any substance accept H^+ is called a BASE

Acidosis $\leftarrow 7.35 - 7.4 \rightarrow$ Alkalosis



coma and death



tetany and convulsions

Renal regulation of pH

Kidney takes several hours to several days to regulate PH. It constitutes the third line of defense. They are the most powerful and most efficient buffering mechanism.

The kidney respond to metabolic acidosis by increasing H^+ secretion into urine and by enhancing bicarbonate re- absorption.

$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$
Carbonic acid $\rightleftharpoons H^+ +$ Bicarbonate (base)

Weak acids partially dissociate and forms basis of the physiological phenomenon called **buffering system**.

pH

There are three major sources of H^+ in the body:

1- The catabolism of proteins that contain amino acids containing phosphorus and sulfur which release phosphoric and sulfuric acids into the extracellular fluids.

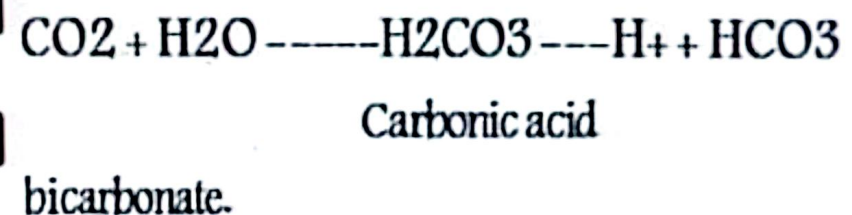
These acids dissociate into

H^+ and anions (Phosphate or sulphate).



Renal Regulation of pH

2- Many organic acids such as fatty acids and lactic acid also produce H^+ in their reactions. The major source of H^+ is from metabolically produced CO_2 .



3- In the lung, elimination of the produced H^+ takes place. However in case of impairment of lung function H^+ accumulates in the blood and so, it must be eliminated through the kidney in order to maintain the H^+ balance.

Renal regulation of pH

1. H^+ is secreted in almost all parts of the renal tubules. In both instances H^+ to be secreted is formed as a result of H_2CO_3 dissociation in the cells.

2. In PCTs: 90% of H^+ is secreted and entirely buffered by the filtered bicarbonate.

Maximal acidifying power up to 6.9.

H^+ secretion occurred in exchange with Na^+ only.

It is not under hormonal control.

Carbonic anhydrase enzyme is essential. Reabsorption of the filtered bicarbonate (the alkali reserve) is achieved.

Renal regulation of extracellular H^+ ions

3. In DCTs and CDs: 10% of H^+ is secreted and buffered by phosphate buffer and ammonia.

Maximum acidifying power is 4.5.

H^+ secretion occurred in exchange with K^+ and by hydrogen pump.

It is under control of aldosterone hormone.

Carbonic anhydrase enzyme is essential.

Renal regulation of extracellular H ions

The excretion of H^+ via the kidney increase or decrease according to the total body production from H^+ .

The ability of the kidney to excrete H^+ depends upon the tubular H^+ secretion and upon the buffers in urine.

The buffering system in urine are:

1- Phosphate buffer system:



2- Ammonia buffer system:



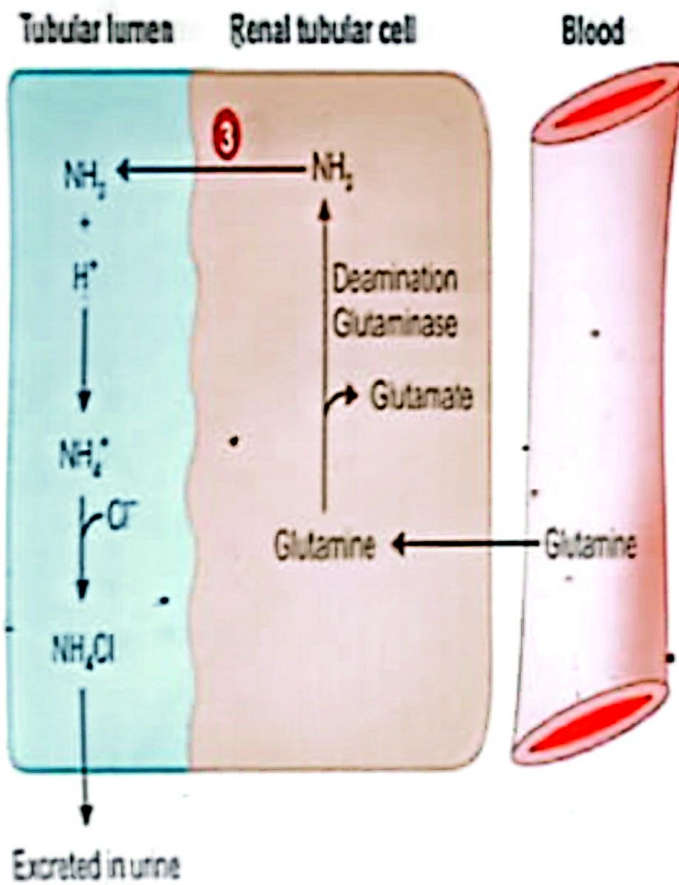
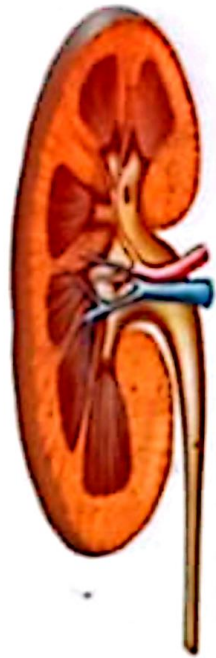
Renal regulation of extracellular H ions

There is a major difference between the HPO_4 which has been filtered and not reabsorbed

and the NH_3 which has been formed in the tubular cells (ammoniogenesis) by deamination of certain amino acids (glutamine), and then NH_3 diffuses into the lumen (lipophilic).

The more H^+ in the lumen, the greater the conversion of NH_3 to NH_4 .

Increased secretion of H^+ automatically inducing an increase excretion of NH_3 in the form of NH_4 .



Acid Base Balance - Renal Mechanism

Urinary phosphate buffer system

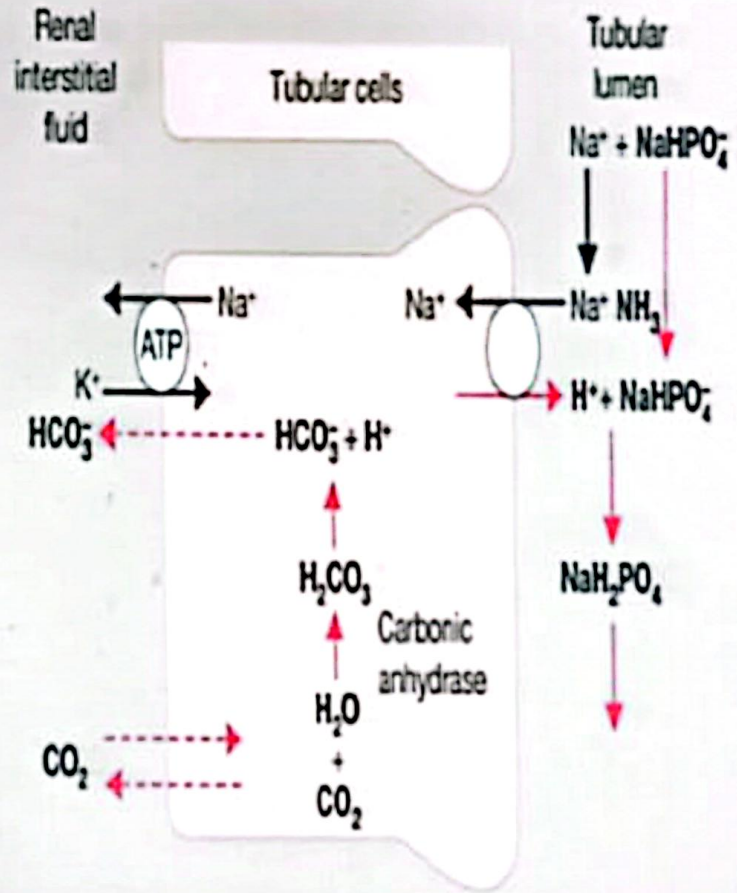


Image courtesy of Guyton & Hall⁴

Effects of renal failure (Uremia)

The respiration becomes deep and rapid in an attempt to compensate for the metabolic acidosis.

Death may occur when the pH of the blood falls to about 6.8 or lower.

Anemia also occurs in chronic renal failure because of severe damage of the kidney and a sharp decrease of erythropoietin which leads to diminished red blood cell production

Effects of renal failure (Uremia)

Renal failure is usually associated with osteomalacia because vitamin D must be converted in the kidney to 1,25-dihydroxycholecalciferol before it is able to promote calcium absorption from the intestine.

Therefore, serious damage to the kidneys greatly reduces the availability of calcium to the bones.

Effects of renal failure (Uremia)

The effect of **renal failure or end stage renal disease (ESRD)** on the body fluids may cause the following effects:

- 1- **Generalized edema** due to water and salt retention.
- 2- **Acidosis** due to failure of the kidney to get rid of the body from the normal acidic products.
- 3- High concentration of non-protein nitrogenous substances such as **urea, creatinine, and uric acid**.
- 4- High levels of other products such as phenols, guanidine bases, sulfates, phosphates, and potassium ions.

Renal regulation of extracellular H⁺ ions

All the above effects lead to a condition called **uremia** because of the high concentrations of normal urinary excretory products that collected in the body fluids (mainly in the blood).

After a week or more of renal failure, the sensorium becomes clouded, and the patient soon progresses into a state of coma probably due to acidosis.

Dialysis of patient with the artificial kidneys

Peoples suffering from, ESRD or permanent renal failure or total removal of kidneys in need of dialysis or kidney transplant or artificial kidney to stay alive.

The basic principle of the artificial kidney is to pass the blood of the patient through very minute blood channels bounded by a thin membrane

Effects of renal failure (Uremia)

On the other side of the membrane is a dialyzing fluid into which unwanted (harmful) substances in the blood such as urea, creatinine could easily pass by diffusion.

In normal person, the urea level is about 26 mg/dl and creatinine is 1 mg/dl.

In uremic person the plasma levels are 200 mg/dl and 6 mg/dl, respectively.

After passing the uremic blood in the artificial kidney, these levels return back to normal and the person becomes in a good health

The artificial kidney

